

Replication package

Diversification, Growth, and Stability in Local Fishing Economies Kyumin Kim and Matthew N. Reimer ·
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Everything in the paper is built from four public CSV files by one command:

```
./run_all.sh # ~15 min; add RUN_BAYES=1 for Table S6 (~20 min)
```

That runs the Stata stage, then the R stage, and writes every figure into `output/figures/` and every saved estimate into `output/estimates/`. If Stata is not on your path, pass it: `STATA=/path/to/stata-mp`
`./run_all.sh`.

Prefer to drive it from Stata itself? Change directory into this folder first — every path in the package is relative to it:

```
cd "/path/to/replication_package"  
do code/stata/00_master.do
```

`00_master.do` also finds the root on its own if you launch it from anywhere inside the package, and stops with an instruction rather than a cryptic error if it cannot.

1. What you need

	Version used	Notes
Stata	17.0 MP	version 17 set in every do-file
R	4.5.3	<code>install.packages("renv"); renv::restore()</code>
JAGS	4.3.2	only for Table S6

Stata packages (`sfcross`, `parwest`, `mat2txt`, `estout`) are **bundled** in `code/ado/` and take precedence, so no SSC access is required. `00_master.do` still checks and names anything missing.

JAGS on Apple Silicon. The CRAN binary of `rjags` looks for `/usr/local/lib/libjags.4.dylib` while Homebrew installs to `/opt/homebrew/lib`, and fails with `Library not loaded`. Build from source:

```
install.packages("rjags", type = "source",  
  configure.args = "--with-jags-include=/opt/homebrew/include/JAGS --with-jags-  
    lib=/opt/homebrew/lib")
```

2. Data

All inputs are public and ship in `data/raw/`. Nothing writes to that folder.

File	Source
akfish-data-CFECpermits.csv	Alaska Commercial Fisheries Entry Commission permit records
employment-by-industry-community-and-year.csv	Alaska Local and Regional Information (ALARI)
employment-summary-data-by-community-and-year.csv	ALARI
Total_pop.csv	Alaska Dept. of Labor population estimates

Study period 2000–2016; analysis sample 177 communities.

3. What produces what

Exhibit	Produced by
Table S1	summary of data/derived/master_local_fish_max.dta
Table S2, Table S5	code/stata/21_ehdf_nmrt.do
Table S3	code/stata/20_ehdf_main.do
Table S4	code/stata/22_robustness_hhi.do
Table S6	code/stata/39_panel_export_for_bayes.do → code/R/14_tableS6_bayesian.R
In-text estimates	code/stata/26_hdf_mrt.do
Fig. 1	code/R/15_fig1_ak_overview.R
Fig. 2	code/R/13_fig2_sector_composition.R
Fig. 3a, 3b	code/R/11_fig3_frontier_concept.R
Fig. 4, Fig. S.2	code/R/12_fig4_figS2_frontier_quantiles.R
Fig. 5	code/R/16_fig5_mrt_marginal_panels.R
Fig. 6	code/R/17_fig6_mrt_substitution.R
Fig. S.1	code/R/10_figS1_diversification_trend.R

Stata tables are read from logs/00_master.log ; the estimates behind them are also saved under output/estimates/.

4. Execution order

Range	Stage
01 – 12	Build master_local_fish_max.dta from the raw CSVs. Order matters.
20 – 26	EHDF estimation, marginal effects, MRT grids. Re-runnable once the master file exists.

Range	Stage
30 – 39	Two-period panel. Only Table S6 uses it.
10 – 17 (R)	Figures and the Bayesian check. Read only what Stata leaves behind.

To run the stages separately:

```
export REPLICATION_ROOT="$PWD"
stata-mp -b do code/stata/00_master.do
Rscript code/R/00_run_all_R.R
```

Stata exits 0 even on error, so `run_all.sh` greps the log for `r(###)`; and fails the run itself if anything appears.

5. Notes for a replicator

The estimation sample is 177, not 179. The `keep if` at the top of each estimation file leaves 179 rows; `sfcross` then drops rows with any missing regressor. `12_build_master.do` asserts 177 so a later edit cannot change the sample silently.

R reads its coefficients from Stata. `24_export_coefficients.do` writes `output/estimates/ehdf_spec3_coefficients.csv`, which the frontier-quantile figures read. Nothing is transcribed by hand.

Table S6 is MCMC. `set.seed(123)` is set, but JAGS is not bit-identical across platforms. Judge agreement within Monte Carlo error: posterior means and SDs moving in the second or third decimal is expected; signs and credible-interval coverage should match. Across four independent runs here, the population coefficient came out 2.772, 2.772, 2.779 and 2.784 against the published 2.775.

One earlier specification is carried along. `25_marginal_effects.do` also produces a legacy version of the frontier estimates that the paper does not report.

6. Layout

<code>run_all.sh</code>	single entry point
<code>config/</code>	<code>paths.do</code> , <code>paths.R</code> -- no absolute paths anywhere
<code>data/raw/</code>	the four public CSVs (inputs; never written to)
<code>data/derived/</code>	built by code
<code>code/stata/</code>	<code>00_master.do</code> + 29 numbered do-files
<code>code/R/</code>	<code>00_run_all_R.R</code> + 8 figure/table scripts
<code>code/ado/</code>	bundled Stata packages
<code>output/</code>	estimates, tables, figures
<code>logs/</code>	<code>00_master.log</code> and one log per R script

7. Licence

Everything here — code, data, and outputs — is released under CC BY 4.0; see [LICENSE](#) . The four raw CSVs are public records of the State of Alaska, redistributed unmodified so the package runs without any external download.

Run on macOS 15 (Apple Silicon), Stata/MP 17.0, R 4.5.3, JAGS 4.3.2.